



Q1. Write a program that uses a while loop to calculate the sum of the first 50 natural numbers?

Q2. Write a program that uses a while loop to find the factorial of a given number?

- num = 6

Q3. Write a program using a while loop that prints numbers from 1 to 10?

Q4. Write a while loop to print the first 20 even numbers?

Q5. Write a program that uses a while loop to find the sum of the digits of a given number?

- num = 54321

Q6. Write a program that reverses a number using a while loop?

Q7. Create an infinite while loop that asks the user to enter a number. If the number is negative, the loop breaks; otherwise, it prints the square of the number?

Q8. write a loop statement to print the following series:

- output : 10,20,30,40,...,300
- append them to a list

Q9. write a while loop statement to print the following series:

- output : 105,98,91,...,7
- append them to list.

Q10. write a program to print first 10 natural numbers in reverse order using while loop?

Q11. write a program to print table of a number entered from the use?

Q12. Write a Python program that repeatedly asks the user to enter a number and prints "Hello" for each input. The program should stop when the user enters a string (non-numeric value)?

Q13. "Write a Python program to print the following sequence of numbers in a single column:

- output : 1,2,3,4,5,6,7,8,9,10,9,8,7,6,5,4,3,2,1

Q14. Write a program that creates a list of names and prints the total number of names in the list using the len() function?

15. write a program to enter the number till the user wants and at the end it should display the sum of all the numbers entered?



- Q16.** create an empty list. Continuously ask the user to enter their favorite movies until they type "done." Use the append() method to add each movie to the list?
- Q17.** write a program to accept 10 numbers from the user and display the largest and smallest number?
- Q18.** write a program to print all the characters in string "PYTHON" using while loop?
- Q19.** write a python program to print only the odd numbers from the given list using while loop?
- list = [23,45,25,32,46,33,71,90].
- Q20.** write a python program to get the following output?
- 1--- 49
 - 2--- 48
 -
 - 49--- 1
- Q21.** Write a program that uses a while loop to check if a given string is a palindrome?
- str = "madam"
- Q22.** Write a program using a while loop to print all the factors of a number entered by the user?
- Q23.** Write a program to find the least common multiple (LCM) of two numbers using a while loop?
- Q24.** Write a program using a while loop to print a right-angled triangle of asterisks (*)?
- Q25.** Create a program that uses a while loop to count the number of vowels in a string entered by the user?
- Q26.** Write a program that uses a while loop to find the average of positive numbers entered by the user until they enter a negative number?
- Q27.** Create a program that uses a while loop to count down from a user-specified number to zero, printing each number?
- Q28.** Create a program that uses a while loop to count the number of digits in a number entered by the user?
- Q29.** Create a program that asks the user for a string and a number, then uses a while loop to print the string the specified number of times?

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Q1. Write a Python function to find the sum of all elements in a list.

Q2. Write a Python program to count the occurrence of each element in a list.

Q3. Write a Python code to generate a list of squares for numbers from 1 to 10.

Q4. Write a Python function that:

- Takes two lists as input.
- Merges the two lists into a single list, alternating elements from each list.
- Returns the merged list.

Q5. Given a list of numbers `numbers = [1, 2, 3, 4, 5, 6, 7]`, write a function to remove multiple elements from a list?

- to remove 3, 5 and 7

Q6. Given a list of integers `[10, 20, 30, 20, 20, 30, 40]`, write a function to remove duplicates. Generate another list, which contains only the unique elements.

Q7. Given 3 digits 4, 5, and 6. Find all the possible combinations of these digits.

Q8. Write a Python program to create a set of even numbers from 1 to 100.

Q9. Write a Python program to find the union of two sets using the `union()` method?

Q10. Given a list of integers, write a function that partitions the list into two parts: one for odd numbers and one for even numbers.

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Q18. covert the following loop into for loop :

- `x = 4`
- `while (x<=8)`
- `print(x*10)`
- `x+=2`

Q19.Python program to check the validity of password

Q20. Create a program that prints each element of a li

Q21. Write a program that prints all numbers between

Q22. Write a program that accepts a string from user a
letters using a for loop?

Q23.Write a program that extracts all digits from a giv

Q24. Given a list of names, write a program that creat
lengths are values using a for loop?

- `names = ["Alice", "Bob", "Charlie", "David", "`

Q25.Accept two strings from user, write a program th
duplicates) using a for loop?

Q26. Write a program that generates a list of n random

Q27.Write a Python program to reverse the order of cl
built function?

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Q1. write a program if a list contains a palindrome of e

- `l = [1,2,3,2,1]`

Q2. write a program to check whether 13 is a prime nu

Q3. Python program to convert the month name to a nu

Q4. Take value of length and breath from the user and

Q5. Take two numerical values from user and print gre

Q6. Calculate the percentage of class attend and say w
exam?

- `working_days = 280`

Q7. write a program to accept percentage from the use

- `< 40 -- Failed`
- `<55 & >=40 -- Fair`
- `<65 & >=55 -- Good`
- `greater or =65 -- Excellent`

Q8. Accept three sides of a triangle from user and che
scalene triangle?

- note :

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Q11. Write a program that takes employees salary as an input and calculates their bonus based on their years of service . The program should be as follows:

- If the years of experience is less than 5 : print(0)
- if the years of experience is more than 5 : calculate the bonus and print it credited.

Q12. write a program to accept two numbers from the user and calculate their sum using a function accordingly?

Q13. write a program to accept the three sides of a triangle and check if it is a right-angled triangle.

Q14. create a program that calculates the library charge for a book borrowed and not returned. The charge structure is as follows:

- For exactly 5 days: Rs 2 per day
- For 6 to 10 days: Rs 3 per day
- For 11 to 15 days: Rs 4 per day
- For more than 15 days: Rs 5 per day

Q15. Write a program to check whether a person is eligible for a driving license or not?

- criteria for voting : more than or equal to 18 - eligible
- criteria for getting driving license: more than or equal to 21 - eligible

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- `l = [1, "abc", 1, "abc"]`

Q25. Write a program that prompts the user to enter the age of a person and whether they are:

- A child (0-12)
- A teenager (13-19)
- An adult (20-64)
- A senior (65 and above)

Q26. Write a program that asks the user to enter a day of the week: weekend (Saturday or Sunday) or a weekday?

Q27. Write a function that calculates the factorial of a number.

Q28. Create a program that takes a number and prints its square and cube.

Q29. create a program to accept the kilometers covered by a person and calculate the cost of the journey.

- First 10 km --- rs 11/km
- Next 90 km --- rs 10/km
- After that --- rs 9/km

Q30. Accept the market price from the user and calculate the amount to be paid after a discount of 20%.

- $\text{amount to be paid} = \text{market_p} - (\text{market_p} * \text{discount})$
- $\text{market_p} > 10,000$ --- discount = 20%

